

The submitted manuscript proposes a weak-value extraction strategy without a meter state or a weak interaction. The authors show that complex weak values of two-path projector observables can be reconstructed from output-port intensity measurements of a generalized Mach-Zehnder interferometer, combined with controlled phase shifts. The manuscript further provides an experimental demonstration of the scheme using neutron-interferometry.

The paper has several real strengths. The central algebra is simple and clean. The neutron interferometer experiment is attractive. The elimination of meter states and weak interactions is a practical contribution. My judgment is that the manuscript is interesting and potentially publishable, but requires some revisions.

1. Clarification of the new insight

The manuscript would benefit from a clearer statement of its specific novelty. The possibility of reconstructing weak values without conventional weak measurements has been discussed previously, for example by Johansen. The meaningful idea of this work appears to be the more specific interferometric implementation: the complex weak values of path projectors are obtained from controlled phase shifts, output-port intensity measurements, and intensity corrections, and the scheme is demonstrated in a neutron Mach-Zehnder interferometer. I encourage the authors to clarify this distinction in the Introduction and Discussion.

2. Notation and normalization of the intensities

Intensities should be clarified in the contrast-correction part of the Methods section. In Eq. (6) and the surrounding equations, $I_{\pm}(\delta)$ appears to denote a physical count rate, since it includes the proportionality factor A . However, in Eq. (17) and the following ones, $I_{+}(\delta)$ is multiplied by A again, which would be dimensionally inconsistent if $I_{+}(\delta)$ still denotes the same count rate. It seems that $I_{+}(\delta)$ here should be replaced by $p_{+}(\delta)$ for ideal probabilities.

3. Resource comparison

The practical resource advantage of the protocol should be discussed more explicitly. In this scheme, the weak values are reconstructed from several intensity measurements, rather than read out directly from a meter shift. Although the absence of ancillary meter states and weak interactions is experimentally appealing, it is not obvious from the current manuscript how the required resources compare with standard qubit tomography or conventional weak-measurement protocols. A quantitative comparison of measurement settings, acquisition time, statistical uncertainty, and experimental complexity would strengthen the manuscript.

4. Treatment of the contrast factor

The treatment and interpretation of the contrast factor C should be clarified. If part of the contrast loss reflects genuine decoherence or an effectively mixed input state, then correcting the data back to an ideal pure-state model may no longer be physically justified. I suggest that the authors clarify the physical meaning and valid range of C , and explain under what assumptions the correction procedure remains reliable.

5. Overstatement of the nonclassicality claim

The manuscript argues that, because no anomalous weak values are observed in the balanced configuration, this configuration does not exhibit nonclassical behavior. This seems too strong. The absence of anomalous real weak values only shows that the particular witness considered here is not triggered; it does not exclude all possible forms of nonclassicality. I therefore suggest that the authors soften this statement.